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ARTICLE



MnO₂ nanozyme induced the chromogenic reactions of ABTS and TMB to visual detection of Fe²⁺ and Pb²⁺ ions in water

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ABSTRACT

In this study, we used a simple and rapid colourimetric reaction for visual sensing of Fe²⁺ and Pb²⁺ ions in water by employing nano-MnO₂ as a natural oxidase mimic to respectively catalyse ABTS and TMB in citrate-phosphate buffer solution (C-PBS) at 25°C and pH 3.8. It was found that nano-MnO₂ possessed highly oxidase-mimicking activity with the *K_m* values of 0.030 and 0.027 toward ABTS and TMB, respectively, indicating TMB had a stronger affinity on nano-MnO₂ than ABTS. Interestingly, the presence of 0.01 mmol·L⁻¹ Fe²⁺/Pb²⁺ ion was able to significantly down-regulate the activity of MnO₂ nanozyme in nano-MnO₂-mediated ABTS reaction processes (*P* < 0.01), which mainly due to the strong adsorption of metal ion toward nano-MnO₂ surface via the electrostatic attractions, thus leading to the passivation and inactivation of MnO₂ nanozyme catalytic activity. Thereinto, Fe²⁺ reacted with multivalent manganese by oxidation-reduction, while Pb²⁺ was specifically adsorbed onto the surface of MnO₂ nanozyme and formed complexes. Notably, only Fe²⁺ ion inhibited the activity of MnO₂ nanozyme-TMB with a detection limit as low as 1.0 µmol·L⁻¹. In MnO₂ nanozyme-ABTS sensing systems, Fe²⁺ and Pb²⁺ ions detection limit of 0.5 and 2.0 µmol·L⁻¹ were, respectively, achieved with a linear response range of 0–0.02 and 0–0.8 mmol·L⁻¹, implying the developed MnO₂ nanozyme-ABTS sensor was potentially applicable for the visual determination of Fe²⁺ and Pb²⁺ ions in water. In the real water samples, MnO₂ nanozyme-ABTS achieved high accuracy (relative errors: 3.4–10.5%) and recovery (96–110%) for respective detection of Fe²⁺ and Pb²⁺ ions. The simple and rapid MnO₂ nanozyme-ABTS sensing systems might provide a practical assay for visual detection of Fe²⁺ and Pb²⁺ ions in the environmental water samples.

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1. Introduction

Currently, nanoscale metal oxides (NMOs) as nanozymes have confirmed to be highly effective in versatile applications [1–3]. Compared with the natural enzymes, enormous attentions have been directed towards the utilisation of NMOs in imitating the

catalytic activity due to their higher stability and lower production cost [4,5]. For example, the intrinsic enzyme-mimicking properties of CuO, CeO₂, Co₃O₄, MnO₂, Mn₃O₄, Fe₂O₃, and Fe₃O₄ have been widely used in bio-sensing, immunoassay, pollutants degradation, and environmental monitoring [6–9]. These applications of nanozymes are based on the high affinity of substrates onto the surface of NMOs, thus altering their catalytic properties [10]. Among these nanozymes, the oxidase-mimicking activity of NMOs especially attracted increasing interest because they avoided the participation of H₂O₂ [11–13].

It is well documented that nanoscale MnO₂ with high stability and recycling properties, as well as remarkable catalytic activity, has become a fascinating nanozyme [11,14]. MnO₂ nanozyme could catalyse the chromogenic reactions of 2,2'-azino-bis (3-ethyl-benzothiazoline)-6-sulphonate (ABTS) and 3,3',5,5'-tetramethylbenzidine (TMB) for the bio-sensing, bio-imaging, single-stranded DNA, and drug delivery [15–17]. For instance, Liu *et al.* revealed that MnO₂ nanozyme could mimic the oxidative activity of oxidase for catalysing the multicolour reaction of TMB via forming blue, green, and yellow products, which had been applied for the detection of glutathione in biological samples [18]. This showed that MnO₂ nanozyme could be proposed to be a possible cost-effective catalyst for the detection of target substance in water [19]. Nonetheless, the enzyme-mimicking activity of nano-MnO₂ has rarely been understood to the visual detection of metal ions in the aquatic environment.

At present, the pollution and hazards of metal ions on wild species and human have increasingly drawn focus worldwide. This is due to they have been extensively applied to the industrial fields, inevitably leading to the release of metal ions into the natural water environment [20]. Notably, accumulation of toxic metal ions higher than hazardous levels can cause acute health problems to most aquatic biota [21]. Therefore, simply and rapidly recognising metal ions is of great significance in the aqueous environment to protect aquatic biota and human health. It is well known that inductively coupled plasma mass spectrometry (ICPMS) could be used to test the metal ions in water; however, the method was costly and time-consuming. Recently, MnO₂ nanozyme-mediated colourimetric reactions have proven to be highly effective for sensing applications because it enables rapid visual detection without the aid of any advanced instruments [17,18,22]. Hence, it is fascinating that whether metal ions can adsorb and deposit onto the surface of MnO₂ nanoparticle to switch its oxidase-mimicking activity with a linear relationship between the concentration of metal ions and the activity of MnO₂ nanozyme [10,23].

In this study, the oxidase-mimicking activity of nanoscale MnO₂ was, respectively, assessed by inducing the chromogenic reactions of ABTS and TMB. In particular, MnO₂ nanozyme was utilised to the visual detection of metal ions in water by the colour variations of reaction systems, based on the fact that metal ions tend to adsorb and deposit onto the surface of nano-MnO₂ as a result of electrostatic attractions, subsequently altering the catalytic activity of MnO₂ nanozyme. Finally, MnO₂ nanozyme-ABTS sensing systems were selectively utilised for the visual detection of Fe²⁺ and Pb²⁺ ions in the real water samples to confirm its potential applications. Our findings might provide a simple and rapid assay for the visual detection of Fe²⁺ and Pb²⁺ ions at the real water samples using MnO₂ nanozyme-ABTS sensor.

2. Experimental

2.1. Chemicals and materials

All reagents used were analytical grade unless otherwise indicated. Nanoscale MnO_2 ($\geq 99.9\%$) was procured from DK Nano Technology Co., Ltd. (Beijing, China). The characteristic of nano- MnO_2 is shown in Fig. S1 (Supplementary Information, SI). ABTS (CAS: 30,931–67-0) and TMB (CAS: 54,827–17-7) were obtained from Energy Chemical Technology Co., Ltd (Shanghai, China). Metal sulfates (*i.e.* $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$, $\text{Al}_2(\text{SO}_4)_3$, CuSO_4 , MgSO_4 , ZnSO_4 , PbSO_4 , and $\text{MnSO}_4 \cdot \text{H}_2\text{O}$) were purchased from Shanghai Aladdin Bio-chem Technology Co., Ltd. Fresh metal ions stock solutions ($100 \text{ mmol} \cdot \text{L}^{-1}$) were freshly prepared in Milli-Q ultrapure water ($18.2 \text{ M}\Omega \cdot \text{cm}$) and stored at 4°C . The buffer used in the experiments was a citrate-phosphate buffer solution (C-PBS: $10 \text{ mmol} \cdot \text{L}^{-1}$ citric acid and $10 \text{ mmol} \cdot \text{L}^{-1}$ Na_2HPO_4 , pH 3.8), which could maintain pH value and ionic strength of all the reaction solutions.

2.2. Activity measurement of nano- MnO_2

To investigate the oxidase-mimicking activity of nano- MnO_2 , the nanoparticle was performed in 10 mL of a C-PBS ($10 \text{ mmol} \cdot \text{L}^{-1}$) at 25°C and pH 3.8 containing $0.01 \text{ mg} \cdot \text{mL}^{-1}$ nano- MnO_2 and $1.0 \text{ mmol} \cdot \text{L}^{-1}$ chromogenic substrate (ABTS or TMB). After nano- MnO_2 and chromogenic substrate had been mixed thoroughly, kinetic measurements of the oxidase-like reactions of nano- MnO_2 were conducted using a UV-Vis spectrophotometer (Shimadzu, UV-2550) in time course mode at 420 nm for ABTS and at 652 nm for TMB in a quartz cuvette with a 1 cm light path. The mixed solution was monitored every 20 s for 3 min by recording the change of absorbance values. One unit of nano- MnO_2 activity ($\text{U} \cdot \text{mL}^{-1}$) is defined as the amount of nanozyme that causes an absorbance change at 420 nm/652 nm in C-PBS ($10 \text{ mmol} \cdot \text{L}^{-1}$) containing $1.0 \text{ mmol} \cdot \text{L}^{-1}$ ABTS/TMB per minute at 25°C and pH 3.8. Thus, the oxidase-like activity of nano- MnO_2 was calculated by the rate of absorbance change. For comparison, experimental nano- MnO_2 -free C-PBS was used as a blank control. All experiments were performed in triplicate.

2.3. Kinetic analysis

Kinetic experiments were conducted in a 50 mL flask at 25°C and pH 3.8 using $1.0 \text{ mmol} \cdot \text{L}^{-1}$ ABTS/TMB with varying concentrations of nano- MnO_2 (0.002 – $0.03 \text{ mg} \cdot \text{mL}^{-1}$) and a fixed concentration of nano- MnO_2 ($0.01 \text{ mg} \cdot \text{mL}^{-1}$) with varying concentrations of ABTS/TMB (0.005 – $1.0 \text{ mmol} \cdot \text{L}^{-1}$) in 10 mL C-PBS ($10 \text{ mmol} \cdot \text{L}^{-1}$). The absorbency variation of reaction systems was recorded at 420 nm for ABTS and at 652 nm for TMB with 20 s interval at kinetics mode for 3 min incubation using UV-Vis spectrophotometer. The kinetic parameters K_m and $V_{\max}$ were obtained by the Lineweaver-Burk plot method as follows: $1/V = (K_m + [S]) / (V_{\max} \cdot [S])$, where V is the initial reaction velocity, $[S]$ is the substrate concentration, K_m is the Michaelis constant, and $V_{\max}$ is the maximal velocity of reaction. All experiments were performed in triplicate.

2.4. Influence of pH and temperature on nano-MnO₂ oxidase-mimicking activity

Similar experimental procedures to those described above were used to evaluate the effects of pH and temperature on the oxidase-mimicking activity of nano-MnO₂. The reactions were performed at different pH (2.0–10.0) and a wide range of temperature (10–90°C). For pH-dependent of reaction, 10 mL of 10 mmol·L⁻¹ C-PBS containing 0.01 mg·mL⁻¹ nano-MnO₂ was mixed with 1.0 mmol·L⁻¹ ABTS/TMB and the temperature was maintained at 25°C. For temperature-dependent of reaction, 10 mL of 10 mmol·L⁻¹ C-PBS containing 0.01 mg·mL⁻¹ nano-MnO₂ was mixed with 1.0 mmol·L⁻¹ ABTS/TMB at pH 3.8. All experiments were performed in triplicate.

2.5. Detection of metal ions by MnO₂ nanozyme

To investigate the effects of metal ions on the activity of MnO₂ nanozyme, a series of mixtures containing different metal ions (*i.e.* Fe²⁺, Al³⁺, Cu²⁺, Mg²⁺, Zn²⁺, Pb²⁺, and Mn²⁺ ions) and nano-MnO₂ (0.01 mg·mL⁻¹) in 10 mmol·L⁻¹ C-PBS were equilibrated at 25°C and pH 3.8 for 30 min, and subsequently 1.0 mmol·L⁻¹ ABTS/TMB was added to each of the mixtures and detected rapidly with 20 s interval for 3 min incubation using UV-Vis spectrophotometer, with a wavelength of 420 nm/652 nm. All experiments were performed in triplicate.

Based on the above results, MnO₂ nanozyme-ABTS sensing systems for Fe²⁺ and Pb²⁺ ions detection in the real water samples were also assessed, respectively. Ten times diluted real water samples (pond water 1–2 from Anhui Agricultural University campus) in C-PBS (10 mmol·L⁻¹, pH 3.8) were spiked with 0–0.02 mmol·L⁻¹ Fe²⁺ or 0–0.8 mmol·L⁻¹ Pb²⁺, and were then detected using the described MnO₂ nanozyme-ABTS sensor following processes similar to those used for the standard solution. The absorbance at 420 nm was measured using UV-Vis spectrophotometer. In particular, the relative errors and recoveries for the detection of Fe²⁺ and Pb²⁺ ions were calculated in MnO₂ nanozyme-ABTS sensing systems, respectively. Additionally, the concentrations of Fe²⁺ and Pb²⁺ ions in the real water samples were also determined by ICPMS (6300 Series, Thermo Fourier, USA), respectively. All experiments were performed in quintuplicate.

2.6. Statistical analysis

All data were processed with Excel 2010 (Microsoft, Redmond, WA). Each datapoint in the figures and tables represents an average value. The standard deviation in parallel samples is shown in the figures with an error bar.

3. Results and discussion

3.1. The oxidase-mimicking activity of nano-mno₂

To test the oxidase-like activity of nano-MnO₂, ABTS and TMB were, respectively, chosen as the oxidase substrates, and all reaction kinetics were monitored at 420 nm for ABTS and at 652 nm for TMB. The changes in absorbance values over time by nano-MnO₂-catalysed oxidation of ABTS and TMB are shown in Figure 1. Nano-MnO₂ could, respectively, catalyse the oxidation of ABTS (Figure 1(a)) and TMB (Figure 1(b)) to form the green and

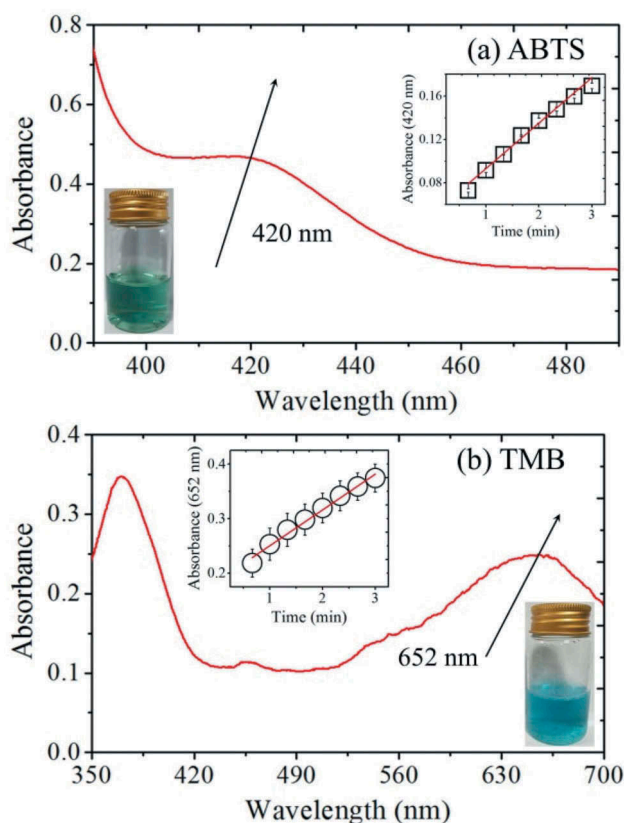


Figure 1. The oxidase-mimicking activities of nano-MnO₂ for catalysing oxidation of substrates. (a) Nano-MnO₂-catalysed absorbency variation of ABTS over time at 420 m; (b) Nano-MnO₂-catalysed absorbency variation of TMB over time at 652 nm. Reaction conditions: [Nano-MnO₂] = 0.01 mg·mL⁻¹, [ABTS] = 1.0 mmol·L⁻¹, [TMB] = 1.0 mmol·L⁻¹g⁻¹, [Temperature] = 25°C, pH 3.8. Error bars represent the standard deviation (n = 3).

blue colour products with a remarkable change in absorption at 420 nm and 652 nm, like laccase-mediated ABTS and TMB chromogenic reactions. Previous reports also indicated that nano-MnO₂ could readily oxidise ABTS and TMB to produce the visual end products of ABTS and TMB at acidic pH, through one-electron transfer processes [17,18,24]. In our study, the oxidase-like activities of nano-MnO₂ were, respectively, 0.041 U·mL⁻¹ for ABTS and 0.065 U·mL⁻¹ for TMB ($R^2 > 0.98$), and nano-MnO₂-free control did not show the oxidase-like activity, demonstrating nano-MnO₂ possessed the oxidase-mimicking activity. Additionally, the reproducibility of nano-MnO₂ was also investigated. The oxidase-like activity of nano-MnO₂ decreased inappreciably as the reaction cycle increased from 0-run to 3-run, suggesting the reusability of nano-MnO₂ as a recyclable oxidase-like.

The role of dissolved O₂ in nano-MnO₂-catalysed oxidation of ABTS and TMB was evaluated by purging the reaction solution with N₂, resulting in a decrease in the oxidase-like activity of nano-MnO₂, and revealing the dissolved O₂ in the catalytic reactions acted as an electron acceptor for MnO₂ nanozyme-substrate to form ABTS and TMB chromogenic products [13,17]. This data is agreement with an earlier report

that indicated the oxidation of substrate via MnO_2 nanosheet and BSA- MnO_2 nanoparticle [18]. Additionally, the stability of nano- MnO_2 in reaction systems was also studied over a one-month storage period. It is noted that the release of Mn^{2+} increased from 0.005 to 0.224 $\text{mg}\cdot\text{L}^{-1}$ as the storage time increased in C-PBS at pH 3.8. Nevertheless, no significant difference in the ABTS and TMB oxidation were detected, implying the capacity of MnO_2 nanozyme to oxidise substrate in reaction systems remain stable. Combining with previous reports on NMOs-based nanozymes, our results demonstrated that nano- MnO_2 possessed a high catalytic activity and stability to, respectively, oxidise the chromogenic reactions of ABTS and TMB.

3.2. Influence of MnO_2 nanozyme concentration on ABTS/TMB oxidation

We further assessed the influence of MnO_2 nanozyme concentration on ABTS and TMB oxidation for a 3-min reaction using UV-Vis spectroscopy. As shown in Figure 2, MnO_2 nanozyme-catalysed oxidation of ABTS and TMB, respectively, showed a distinct absorption peak at the wavelength of 420 nm and 652 nm, and the increase of this absorbance

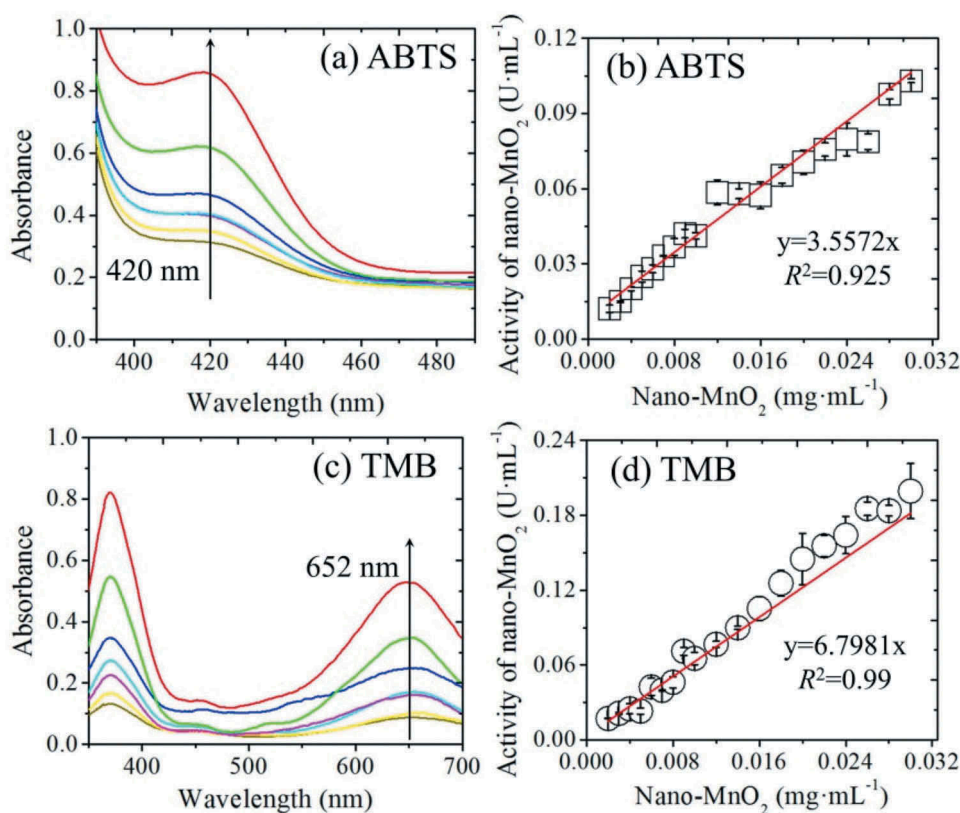


Figure 2. (a, c) UV-Vis absorption spectra of ABTS and TMB reacted with nano- MnO_2 nanozyme for a 3-min incubation; (b, d) Concentration-dependent kinetic studies of the oxidase-like activities of nano- MnO_2 for ABTS and TMB reactions. Reaction conditions: $[\text{Nano-MnO}_2] = 0.002\text{--}0.03 \text{ mg}\cdot\text{mL}^{-1}$, $[\text{ABTS}] = 1.0 \text{ mmol}\cdot\text{L}^{-1}$, $[\text{TMB}] = 1.0 \text{ mmol}\cdot\text{L}^{-1}$, $[\text{Temperature}] = 25^\circ\text{C}$, pH 3.8.

values over MnO_2 nanozyme concentration were obvious resulting from ABTS and TMB oxidation by MnO_2 nanozyme (Figure 2(a,c)). Moreover, the variation of MnO_2 nanozyme activity was also observed by adding different concentrations of nano- MnO_2 . Increasing the concentrations of nano- MnO_2 from 0.002 to 0.03 $\text{mg}\cdot\text{mL}^{-1}$ resulted in a linear increase in the oxidase-mimicking activities of nano- MnO_2 by the oxidation of ABTS and TMB (Figure 2(b,d)). According to the estimated correlation coefficients of nano- MnO_2 concentration and its oxidase-like activity, the apparent pseudo-second-order rate coefficients were, respectively, determined to be 3.56 $\text{U}\cdot\text{mg}^{-1}$ for ABTS and 6.80 $\text{U}\cdot\text{mg}^{-1}$ for TMB ($R^2 > 0.92$). These results demonstrated that increasing the concentrations of nano- MnO_2 facilitated the oxidase-like activities of nano- MnO_2 to catalyse the oxidation of ABTS and TMB.

3.3. Influence of substrate concentration on ABTS/TMB oxidation

For illuminating the catalytic reaction mechanism and gaining the kinetic parameters (K_m and $V_{\max}$), MnO_2 nanozyme-catalysed the steady-state kinetics of 0.005–1.0 $\text{mmol}\cdot\text{L}^{-1}$ ABTS and TMB oxidation were investigated in 10 mL C-PBS for 1 min incubation at 25°C and pH 3.8. The K_m and $V_{\max}$ values were determined from Lineweaver–Burk plots (Fig. S2, SI). In MnO_2 nanozyme-mediated reaction systems, both ABTS and TMB were transformed into the chromogenic products could make the nano- MnO_2 -based nanozyme an intrinsic signal amplifier (Figure 3). Interestingly, the concentration of TMB was in the range of 0.005–0.01 $\text{mmol}\cdot\text{L}^{-1}$, the reaction solution was yellow, while the concentration of TMB was in the range of 0.02–0.04 $\text{mmol}\cdot\text{L}^{-1}$ and over 0.04 $\text{mmol}\cdot\text{L}^{-1}$, the reaction solution showed blue and green colours, respectively, which mainly due to

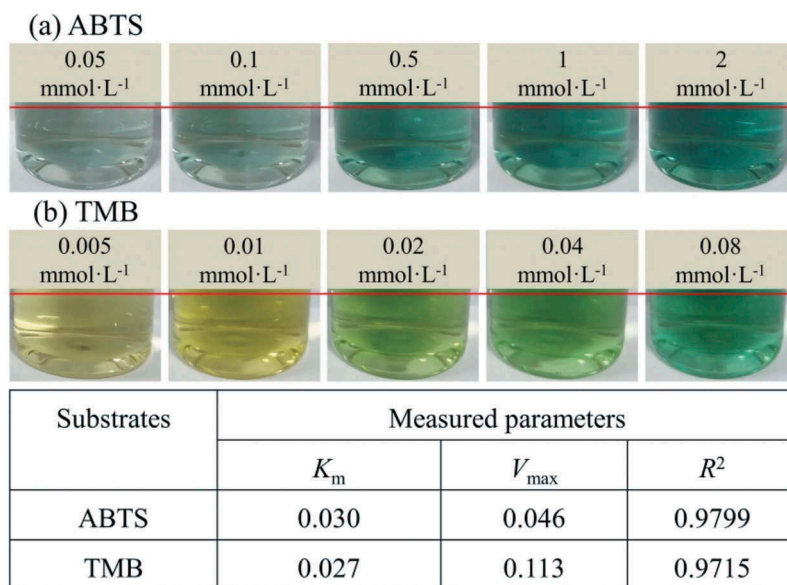


Figure 3. Experimental K_M and $V_{\max}$ values determined for MnO_2 nanozyme-mediated oxidation of ABTS and TMB, respectively. Reaction conditions: $[\text{Nano-MnO}_2] = 0.01 \text{ mg}\cdot\text{mL}^{-1}$, $[\text{ABTS}] = 0.005\text{--}1.0 \text{ mmol}\cdot\text{L}^{-1}$, $[\text{TMB}] = 0.005\text{--}1.0 \text{ mmol}\cdot\text{L}^{-1}$, $[\text{Temperature}] = 25^\circ\text{C}$, $\text{pH } 3.8$.

the formation of TMB diamine and diimine by MnO_2 nanozyme [18,25]. From the kinetic analysis, it was found that the K_m values were, respectively, 0.030 and 0.027 for ABTS and TMB (also listed Figure 3), indicating MnO_2 nanozyme showed a higher affinity towards TMB than ABTS. The $V_{\max}$ values of MnO_2 nanozyme for ABTS and TMB were, respectively, 0.046 and 0.113, revealing MnO_2 nanozyme-TMB reacted faster than MnO_2 nanozyme-ABTS. It is noted that the oxidase-like activity of nano- MnO_2 and the kinetic parameter values were investigated at acidic pH because of limited oxidase-like activity at neutral and alkaline pH.

3.4. Influence of pH and temperature on ABTS/TMB oxidation

pH and temperature were played two vital factors in the oxidase-like activity of nano- MnO_2 . Similarly to the natural oxidase, the catalytic activity of MnO_2 nanozyme was also dependent on pH and temperature [26]. As shown in Figure 4, the effect of pH and temperature on the oxidase-like activities of nano- MnO_2 demonstrated that the catalytic activities of MnO_2 nanozyme decreased from 0.1019 to 0 $\text{U}\cdot\text{mL}^{-1}$ for ABTS and from

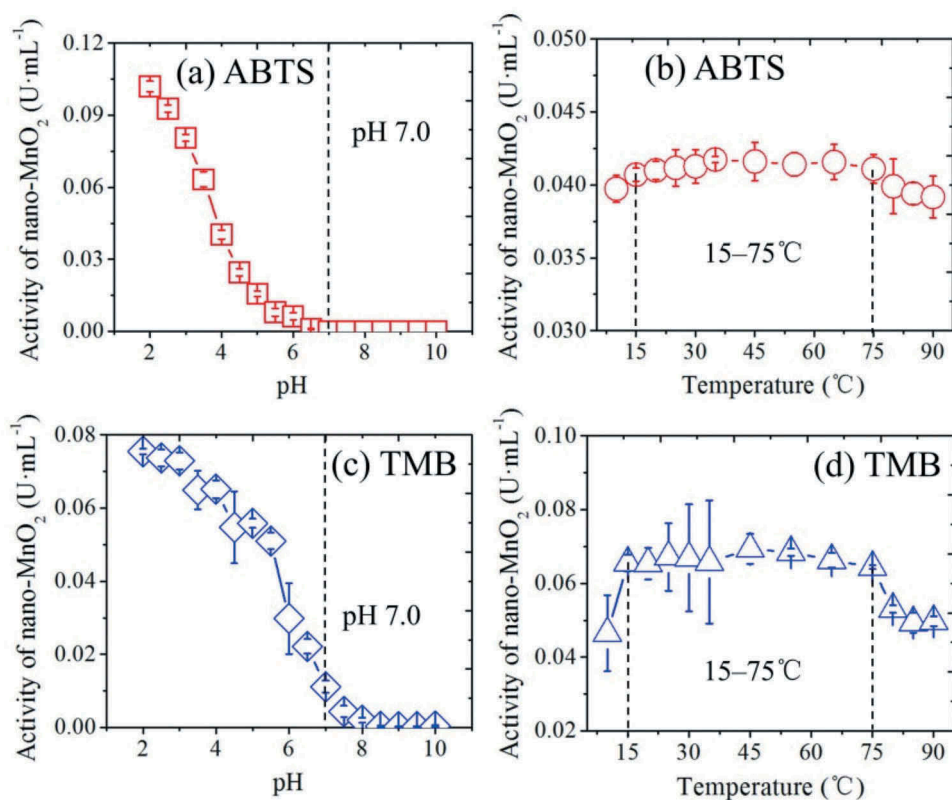


Figure 4. pH- and temperature-dependent the oxidase-like activities of nano- MnO_2 for catalysing the oxidation of ABTS and TMB. (a, c) For pH-dependent of reactions, the temperature was maintained at 25 $^{\circ}\text{C}$; (b, d) For temperature-dependent of reactions, the pH was maintained at 3.8. Reaction conditions: $[\text{Nano-MnO}_2] = 0.01 \text{ mg}\cdot\text{mL}^{-1}$, $[\text{ABTS}] = 1.0 \text{ mmol}\cdot\text{L}^{-1}$, $[\text{TMB}] = 1.0 \text{ mmol}\cdot\text{L}^{-1}$, $[\text{Temperature}] = 10\text{--}90^{\circ}\text{C}$, pH 2.0–10.0. Error bars represent the standard deviation ($n = 3$).

0.0755 to 0.0112 U·mL⁻¹ for TMB with the rise of reaction pH from 2.0 to 7.0, whereas the oxidase-like activities of MnO₂ nanozyme remained stable from 15°C to 75°C. It was found that only 0.006 and 0.030 U·mL⁻¹ of MnO₂ nanozyme activities were retained at pH 6.0 for ABTS and TMB, respectively, while 0.039 and 0.050 U·mL⁻¹ of MnO₂ nanozyme activities were, respectively, retained even at 90°C for ABTS and TMB. Earlier studies also demonstrated that the catalytic oxidation of ABTS and TMB by different nanozymes was found maximum at acidic pH than neutral and alkaline conditions [18,27,28]. These results indicated that the comparative oxidase-like activity of nano-MnO₂ over a broad range of pH and temperature, exhibiting a wide range of pH and thermal stability, unlike natural oxidase was easily denatured and inactivated [28,29]. Based on the above studies, the experimental detecting conditions of MnO₂ nanozyme activity were thus obtained as follows: pH 3.8, 25°C, and 0.01 mg·mL⁻¹ of nano-MnO₂.

3.5. Effect of metal ions on MnO₂ nanozyme-catalysed ABTS/TMB oxidation

Simply and rapidly recognising metal ions is of important significance in the aqueous environment. Previous studies have indicated that BiOI, Fe₃O₄, and BSA-stabilised Pt nanozymes could be used to rapidly detect Hg²⁺, Pb²⁺, and Zn²⁺ ions due to their intrinsic advantage and high stability under harsh conditions [30,31]. In this study, the activities of MnO₂ nanozyme were selectivity evaluated by the various metal ions involving Fe²⁺, Al³⁺, Cu²⁺, Mg²⁺, Zn²⁺, Pb²⁺, and Mn²⁺ ions in 10 mL C-PBS at 25°C and pH 3.8. As shown in Figure 5, the oxidase-mimicking activities of nano-MnO₂-ABTS and nano-MnO₂-TMB were, respectively, 0.039 and 0.068 U·mL⁻¹ in the blank controls (BC, *i.e.* metal ion-free). Compared with BC, the activities of nano-MnO₂-ABTS were significantly inhibited in the presence of Fe²⁺ and Pb²⁺ ($P < 0.01$), whereas the activity of nano-MnO₂-TMB was only suppressed with Fe²⁺ present ($P < 0.05$). The restraining activity of MnO₂ nanozyme with Fe²⁺ and Pb²⁺ ions present occurred likely due to the strong adsorption of Fe²⁺ and Pb²⁺ ions onto the nano-MnO₂ surface via the electrostatic attractions, which impeded the single-electron transfer to the substrate [30,31]. Additionally, we also investigated and compared the influence of Co²⁺, Fe³⁺, Hg²⁺, K⁺, Ca²⁺, Ag²⁺, and Cd²⁺ ions on MnO₂ nanozyme activity. Similarly, there was no significant effect on the activity of MnO₂ nanozyme by them. These results demonstrated that MnO₂ nanozyme might be used to easily and rapidly detect Fe²⁺ and Pb²⁺ ions in aqueous solution.

3.6. MnO₂ nanozyme-based sensor for detecting Fe²⁺ and Pb²⁺

It is possible to derive a quantitative relationship between metal ion concentration and the activity of MnO₂ nanozyme. As shown in Figure 6, a good linear correlation between the oxidase-like activity of nano-MnO₂ and the concentration of Fe²⁺/Pb²⁺ ion was established at 25°C and pH 3.8. The correlation coefficients values ($R^2 > 0.98$) for Fe²⁺ and Pb²⁺ ions were -3.9974 and -0.0472, respectively (Figure 6(a,b)). In particular, increasing the concentration of Fe²⁺/Pb²⁺ ion results in a colour progression from green to colourless in MnO₂ nanozyme-ABTS systems. Similarly, increasing the concentrations of Fe²⁺ ion results in a colour progression from blue to colourless in MnO₂ nanozyme-TMB systems. The correlation coefficients values ($R^2 > 0.96$) for Fe²⁺ ion was -1.6303 (Figure 6(c)).

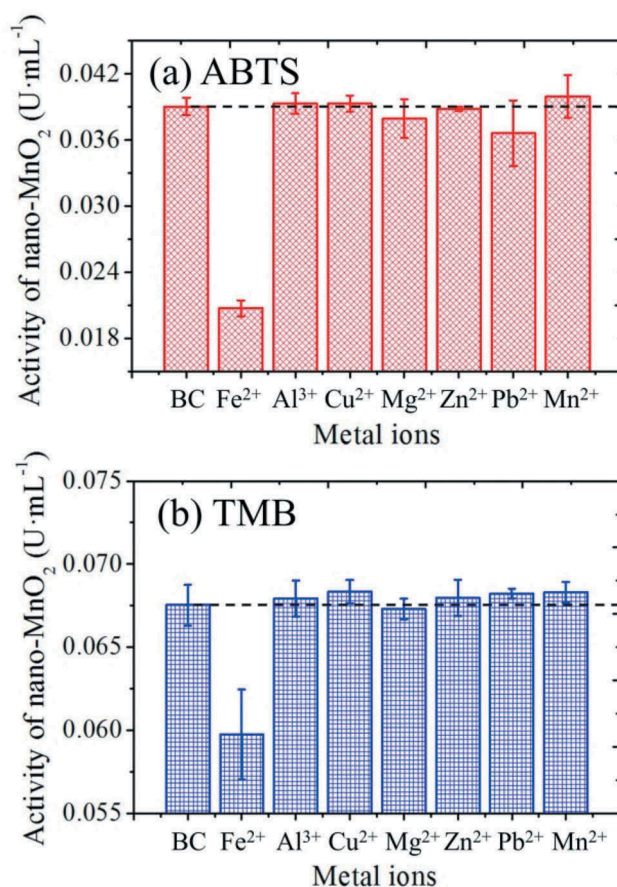


Figure 5. Effect of different metal ions (*i.e.* Fe²⁺, Al³⁺, Cu²⁺, Mg²⁺, Zn²⁺, Pb²⁺, and Mn²⁺ ions) on MnO₂ nanozyme activities by catalysing the oxidation of ABTS and TMB, respectively. Reaction conditions: [Nano-MnO₂] = 0.01 mg·mL⁻¹, [ABTS] = 1.0 mmol·L⁻¹, [TMB] = 1.0 mmol·L⁻¹, [Metal ion] = 0.01 mmol·L⁻¹, [Temperature] = 25°C, pH 3.8. Error bars represent the standard deviation (n = 3).

Hsu *et al.* indicated that BiOI nanocomposite-based sensing systems allowed the detection of Pb²⁺ down to nanomolar quantity [30]. Nawaz *et al.* revealed that cellulose-based fluorescent sensor could be used for the detection of Fe²⁺ ion, and the detection limit for Fe²⁺ ion was micromolar range [32]. In our study, the high sensitivity and wide dynamic range for the detection of Fe²⁺ ion proved that MnO₂ nanozyme allowed for detecting Fe²⁺ ion with a limit of detection of less than 0.5 μmol·L⁻¹, which was lower than the maximum level of Fe²⁺ ion (5 μmol·L⁻¹) in drinking water permitted by the national standards GB 5749–2006 sanitary standard for drinking water in China. Therefore, the MnO₂ nanozyme-ABTS sensor was potentially applicable for visual determination of low-level Fe²⁺ ion in water.

To further explore the possible interactions between metal ion and MnO₂ nanozyme, a differential UV-Vis spectrometry approach was performed [33]. The differential absorbance spectrum (DAS) was calculated by the following equation: $\Delta A_{\text{DAS}} = A_{\text{mixture}} - A_{\text{substrate}} - A_{\text{metal ion}}$, where A_{mixture} , $A_{\text{substrate}}$, and $A_{\text{metal ion}}$ are,

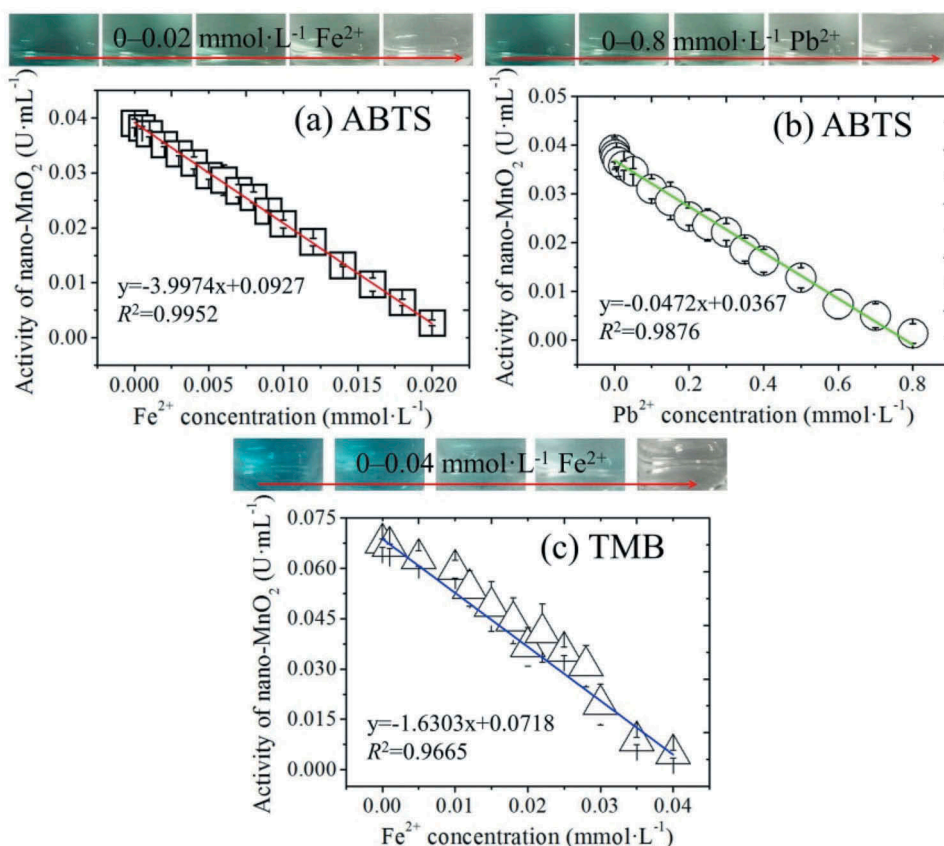


Figure 6. Linear correlation between the oxidase-like activity of nano-MnO₂ and the concentration of metal ion. (a, b) Fe²⁺ (0–0.02 mmol·L⁻¹) and Pb²⁺ (0–0.8 mmol·L⁻¹) ions concentrations-dependent the oxidase-like activities of nano-MnO₂ for catalysing the oxidation of ABTS; (c) Fe²⁺ (0–0.04 mmol·L⁻¹) ion concentrations-dependent the oxidase-like activities of nano-MnO₂ for catalysing the oxidation of TMB. Reaction conditions: [Nano-MnO₂] = 0.01 mg·mL⁻¹, [ABTS] = 1.0 mmol·L⁻¹, [TMB] = 1.0 mmol·L⁻¹, [Temperature] = 25°C, pH 3.8. Error bars represent the standard deviation (n = 3).

respectively, the absorbance wavelength of the mixture solution at 390–490 nm for ABTS and at 350–700 nm for TMB, and the corresponding reference ABTS/TMB and metal ion solution. As shown in Figure 7, the DAS of MnO₂ nanozyme-ABTS systems with Fe²⁺ and Pb²⁺ ions present had an intensive positive peak at 420 nm (Figure 7 (a,b)), while the DAS of MnO₂ nanozyme-TMB systems with Fe²⁺ ion present had two intensive positive peaks at 370 and 652 nm (Figure 7(c)). The variation of DAS implied that Fe²⁺ and Pb²⁺ ions strongly adsorbed to the surface of MnO₂ nanozyme via the electrostatic attractions, thereby leading to the one-electron transfer between the MnO₂ nanozyme and substrate might have been impeded [30]. Notably, Fe²⁺ was adsorbed onto the surface of MnO₂ nanozyme and reacted with multivalent manganese by oxidation-reduction, while Pb²⁺ was specifically adsorbed onto the surface of MnO₂ nanozyme and formed complexes.

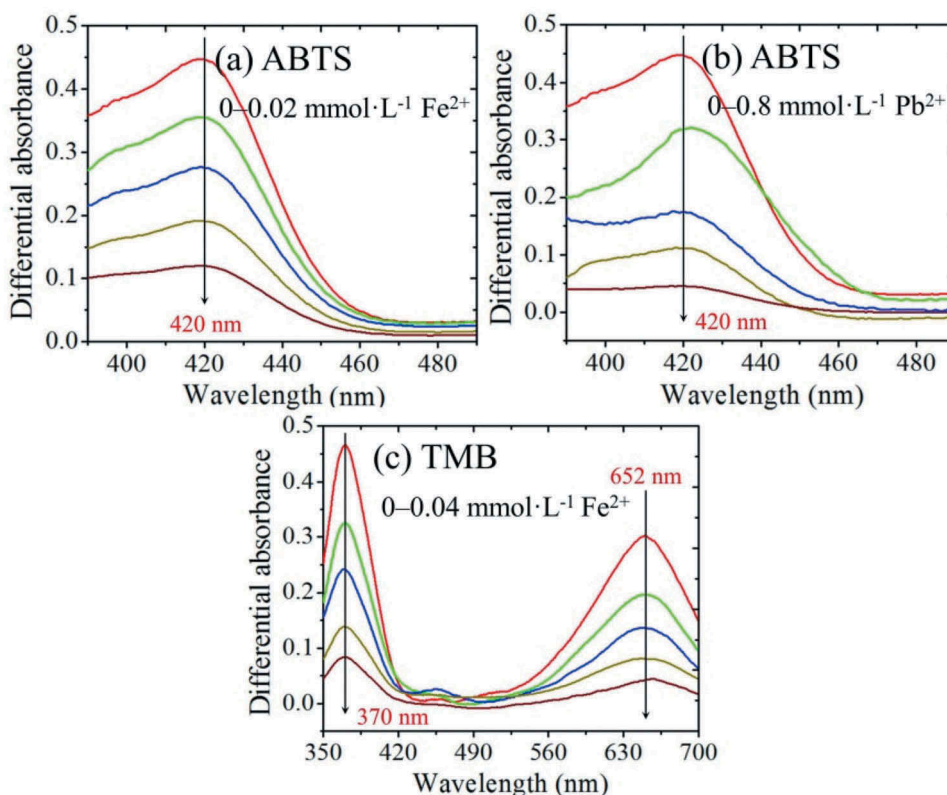


Figure 7. The UV-Vis differential absorbance spectra (DAS) calculated on the basis of the data recorded at different metal ion concentrations. (a, b) Fe^{2+} ($0\text{--}0.02\text{ mmol}\cdot\text{L}^{-1}$) and Pb^{2+} ($0\text{--}0.8\text{ mmol}\cdot\text{L}^{-1}$) ions concentrations-dependent the UV-Vis DAS in MnO_2 nanozyme-ABTS systems; (c) Fe^{2+} ($0\text{--}0.04\text{ mmol}\cdot\text{L}^{-1}$) ion concentrations-dependent the UV-Vis DAS in MnO_2 nanozyme-TMB systems. Reaction conditions: $[\text{Nano-MnO}_2] = 0.01\text{ mg}\cdot\text{mL}^{-1}$, $[\text{ABTS}] = 1.0\text{ mmol}\cdot\text{L}^{-1}$, $[\text{TMB}] = 1.0\text{ mmol}\cdot\text{L}^{-1}$, $[\text{Temperature}] = 25^\circ\text{C}$, $\text{pH } 3.8$. Error bars represent the standard deviation ($n = 3$).

3.7. Detection of Fe^{2+} and Pb^{2+} ions in the real water by MnO_2 nanozyme-ABTS

To verify MnO_2 nanozyme is applicable for the real environmental water samples, spiked-recovery tests with different concentrations of Fe^{2+} and Pb^{2+} ions were, respectively, carried out using the pond water 1 and 2 from Anhui Agricultural University via above-described method. The recovery rates and relative errors were, respectively, calculated as 96–110% and 3.4–10.5% for Fe^{2+} and Pb^{2+} ions spiked in the pond water 1 and 2 (Table S1, SI), implying the reliability and feasibility of the proposed method. The concentrations of Fe^{2+} and Pb^{2+} ions in the pond water 1 and 2 were also determined by ICPMS. As a result, the concentrations of 0.001–0.0206 and 0.04987–0.4013 $\text{mmol}\cdot\text{L}^{-1}$ were, respectively, determined for Fe^{2+} and Pb^{2+} ions using ICPMS, which showed no significant difference compared with MnO_2 nanozyme-ABTS. These results confirmed that the developed MnO_2 nanozyme-ABTS sensing systems could be potentially applicable for easily and rapidly quantising the concentration of Fe^{2+} and Pb^{2+} ions in the real water environment, respectively. However, MnO_2 nanozyme was unable to determine

ultralow-level Fe^{2+} ion and/or low-level Pb^{2+} ion, therefore, a more sensitive method should be developed with a lower detection limit.

4. Conclusions

In our study, nanoscale MnO_2 as a catalyst possessed highly oxidase-like activity at 25°C and pH 3.8, which was confirmed by the chromogenic reactions of ABTS and TMB. MnO_2 nanozyme showed several advantages compared with natural enzymes, for instance, easy preparation, low-cost, higher stability, and better catalytic activity. Hence, we developed MnO_2 nanozyme-substrate systems to selectively detect metal ions in C-PBS and pond water. Results indicated that the presence of $0.01 \text{ mmol}\cdot\text{L}^{-1}$ $\text{Fe}^{2+}/\text{Pb}^{2+}$ ion significantly reduced the reaction activity of MnO_2 nanozyme-ABTS ($P < 0.01$), mainly owing to the strong adsorption of $\text{Fe}^{2+}/\text{Pb}^{2+}$ ion toward nano- MnO_2 , thereby altering the surface properties of MnO_2 nanozyme by different mechanisms. Fe^{2+} and Pb^{2+} ions detection limit of 0.5 and $2.0 \mu\text{mol}\cdot\text{L}^{-1}$ were, respectively, achieved with a linear response range of 0–0.02 and 0–0.8 $\text{mmol}\cdot\text{L}^{-1}$. In the real water samples, MnO_2 nanozyme-ABTS also achieved high accuracy (relative errors: 3.4–10.5%) and recovery (96–110%) for respective detection of Fe^{2+} and Pb^{2+} ions. These findings suggested that MnO_2 nanozyme-ABTS sensing systems might provide a potential application for visual detection of Fe^{2+} and Pb^{2+} ions in the environmental water samples.

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Disclosure statement

No potential conflict of interest was reported by the authors.

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